

Fine-Grained Spatial Scene Reconstruction for Adaptive Memory in Virtual Reality

Edward Scott Carvalho Johnson*
Federal University of Goiás

Bianca Mirtes Araújo Miranda
Digital Metropolis Institute

Arthur Galdino Dangoni
AKCIT

Pedro Elias Engelberg Silva Borges
AKCIT
Federal University of Goiás

Federal University of Rio Grande do Norte
Diogo Fernandes Costa Silva
AKCIT
Federal University of Goiás

Federal University of Goiás
Arlindo Rodrigues Galvão Filho
AKCIT
Federal University of Goiás



Figure 1: Constructive spatial memory reconstruction as iterative scene editing. From an initial panoramic scaffold (left), the remembered environment is progressively rebuilt through explicit insertion and adjustment of spatial cues, illustrating reconstruction as an active and editable process rather than passive replay.

ABSTRACT

Human spatial memory is reconstructive rather than veridical, relying on partial cues, prior knowledge, and iterative refinement. Existing XR memory systems largely emphasize capture fidelity and replay, offering limited support for this constructive process. We present a VR system that frames spatial memory recall as interactive scene reconstruction through explicit editing operations. The system treats a remembered environment as an editable spatial artifact, supporting insertion, movement, cloning, and deletion of scene elements. A hybrid approach combines a 360 panorama as an egocentric contextual scaffold with manipulable 3D objects for fine-grained spatial adjustment, alongside localized panorama editing for correction and suppression of background elements. By operationalizing reconstruction as a sequence of visible edits rather than passive playback, this work provides a foundation for studying how XR tools shape remembered spatial scenes and supports future behavioral evaluation within the XRMemory context.

Index Terms: Spatial memory, Memory reconstruction, Extended reality, Scene editing.

1 INTRODUCTION

Spatial episodic memory, including autobiographical recollection of places and events, depends strongly on contextual structure such as layouts, landmarks, and object-to-object relations. However, human remembering is not a lossless recording. A remembered scene is typically reconstructed from partial details guided by schemas and prior knowledge, which can produce both adaptive generalization and systematic distortions [1, 12]. In XR, this implies that memory playback alone may be insufficient to support recall when

remembered content is incomplete or uncertain.

Recent XR memory work has emphasized immersive capture formats and interaction with recorded experiences, including frameworks for accessing XR memories and re-experiencing past events [7, 10]. Other work highlights that replay alone can be insufficient when contextual cues are missing or when later viewers need interactive access to intent and nonverbal information [9]. At the same time, XR can also shape what is encoded and later recalled, including cases where gradual environmental changes may lead to blended spatial representations [14]. These observations motivate approaches that move beyond capture fidelity and treat remembered content as something that can be actively reconstructed.

Reconstruction operations are rendered explicit and observable, enabling empirical study of how spatial memories are assembled, revised, and distorted through interaction. These processes are difficult to capture with replay- or recognition-based XR memory systems. This paper adopts the framing of constructive spatial memory reconstruction through scene editing. We take the position that an XR memory system can support recall by providing explicit operations for assembling, adjusting, and revising spatial cues, rather than by emphasizing capture fidelity or replay.

In this context, we make three contributions. First, we present an interaction grammar for spatial memory reconstruction in VR, framed as an analytic vocabulary of first-class reconstruction operators, namely insertion, movement, cloning, and deletion, that make the constructive process of recall explicit and observable. Second, we describe a hybrid reconstruction approach that combines a 360 panorama as an egocentric contextual scaffold with manipulable 3D object cues to support fine-grained spatial adjustment. Third, we introduce a localized panorama editing workflow that enables removal and correction of scene elements as part of the reconstruction process. Together, these contributions establish a methodological framework for studying constructive spatial memory in XR through explicit reconstruction actions and provide a concrete system for treating XR memory as an editable spatial artifact.

*e-mail: edward.johnson@discente.ufg.br

2 RELATED WORKS

2.1 Constructive remembering and scene-based recall

Classic and modern accounts converge on the view that remembering is constructive and schema-guided, not a faithful reproduction of experience [1, 12]. Bartlett's schema view explains why recollections often become normalized toward familiar structures, with omissions, substitutions, and regularization that reflect prior knowledge rather than the original input [1]. Schacter frames these distortions as outcomes of adaptive constructive processes that support flexible cognition, including recombining elements across experiences, while also producing predictable errors and false details [12]. In this framing, editing is not an artificial action layered on top of memory, it matches the way reconstruction already proceeds under uncertainty.

A related line argues that episodic recall and episodic future thinking rely on constructing a coherent scene that binds elements into a spatially organized representation [6, 5]. Hassabis and Maguire argue that scene construction is a core component process that supports recollective experience across tasks that require a spatially coherent setting, and that deficits in this process can explain broader impairments in episodic recall and imagination [6]. Byrne et al. further describe how spatial cognition depends on interactions between allocentric and egocentric representations, including transformations between these frames during retrieval and imagery [5]. These perspectives imply two design requirements for XR reconstruction tools. The target artifact should be scene-like rather than propositional, and the system should make iterative revision inexpensive, since reconstruction is incremental and error-prone.

Taken together, these accounts suggest that reconstruction during recall involves a small number of recurring cognitive operations rather than a single replay process. Remembered scenes are incrementally assembled by reinstating missing elements, adjusting spatial relations, externalizing implicit contextual cues, and suppressing details that are inconsistent or misleading. These processes are not arbitrary edits but reflect how schemas guide completion, how spatial frames are transformed during imagery, and how adaptive distortions arise during retrieval [1, 12, 6, 5].

This perspective motivates treating reconstruction actions as explicit operators rather than as incidental interactions. In the system presented here, insertion corresponds to schema-guided completion of missing cues, movement supports iterative refinement of spatial relations across egocentric and allocentric frames, cloning externalizes contextual elements into manipulable representations that can be re-integrated into a scene, and deletion reflects the suppression or correction of misleading details. Framing these actions as first-class reconstruction operators distinguishes recall-oriented interaction from general-purpose scene editing and provides a basis for analyzing how remembered scenes are actively constructed over time.

2.2 Context, spatial representation, and XR memory systems

The encoding specificity principle predicts that recall benefits when retrieval cues reinstate aspects of the original context [13]. Tulving and Thomson emphasize that cues are effective when they overlap with how information was encoded, which makes retrieval sensitive to contextual features rather than only item strength [13]. Spatial memory research further distinguishes egocentric and allocentric contributions to how environments are represented and navigated, suggesting that multiple complementary representations can support recall and reasoning about space [4]. Burgess argues that these representations exist in parallel and combine depending on task demands, motivating systems that can support both view-based context and map-like structure [4].

In parallel, XRMemory-related work emphasizes immersive capture formats and interaction with recorded experiences, defin-

ing XR memories broadly and proposing interaction processes for recalling and managing them [7]. TangibleMoments introduces an interaction vocabulary for working with XR memories and treats memory artifacts as manipulable objects, which supports the idea that recall in XR involves operations over stored spatial media rather than passive replay [7]. TeleAbsence explores interfaces that link tangible objects, scanned spaces, and virtual artifacts to evoke autobiographical recollection, emphasizing that re-experiencing relies on cues and reconstruction rather than literal reproduction [10]. Avagent argues that replay alone can miss critical contextual information and proposes interactive reconstruction mechanisms that preserve intent and nonverbal context for later access [9]. At the same time, XR can shape what is encoded and later recalled, including cases where gradual environmental transitions lead to blended spatial representations [14]. This result motivates treating reconstruction edits as traceable actions and considering how editing operations might influence later recall. Work on memory recognition in VR further suggests that context effects depend on task structure, motivating recall-focused evaluations rather than recognition-only measures [8].

Our approach differs from general 360 authoring and retouching tools in both intent and use context. Conventional panorama editing workflows primarily support localized patching, inpainting, seam correction, and other operations aimed at removing capture artifacts or improving visual fidelity for production. These tools assume a stable ground truth scene and are optimized for visual consistency rather than epistemic uncertainty. In contrast, the localized panorama edits in our system are invoked as part of memory reconstruction, where removal, correction, or alteration reflects uncertainty, reinterpretation, or suppression of remembered elements rather than aesthetic refinement. Furthermore, cloning background elements into manipulable 3D cues links panorama editing directly to spatial scene reconstruction, coupling image-level edits with memory-oriented interaction in ways not supported by conventional 360 authoring pipelines.

2.3 Generative editing and object creation as enabling components

Our implementation uses generative models as enabling components rather than as the primary contribution. Instruction-guided image editing with diffusion models provides a practical mechanism for localized edits [3, 16]. InstructPix2Pix formulates editing as following natural language instructions to transform an input image into an edited output, which aligns with interactive reconstruction workflows where users specify small corrections rather than full regeneration [3]. ControlNet provides a way to condition diffusion models on additional spatial controls such as edges, depth, or segmentation, which is relevant when edits must respect geometric structure and preserve scene consistency [16].

Text-to-3D methods enable turning semantic descriptions into spatialized 3D cues suitable for manipulation in VR [11]. DreamFusion demonstrates that a pretrained 2D diffusion model can act as a prior that drives optimization of a view-consistent 3D representation from text, motivating text-driven or image-driven 3D asset creation when large paired 3D datasets are unavailable [11]. Panorama generation and editing raise additional constraints such as global coherence and seam continuity, motivating the use of panorama-capable diffusion pipelines where available [15]. PanoDiffusion targets wraparound consistency explicitly during diffusion, which is relevant to panorama editing where local changes can introduce boundary artifacts if global continuity is not enforced [15]. These components are replaceable, and the paper's contribution lies in the reconstruction-oriented interaction grammar that integrates them into explicit scene-editing operations.

3 SYSTEM OVERVIEW

Our system treats a memory as an editable spatial scene whose state consists of contextual structure and a set of spatial cues. Reconstruction is expressed through a small set of edit operators that transform this scene state by adding, adjusting, externalizing, or suppressing cues. Users can work in one of two modes, each supporting a different reconstruction workflow.

3.1 Mode A 360 panorama scaffold for egocentric reconstruction

This mode starts from a 360 environment selected from a library or generated from a voice prompt. The panorama provides an egocentric contextual scaffold that reinstates layout and background cues supporting recall [13]. Rather than serving as a static backdrop, the panorama functions as a reference frame against which remembered elements are compared, adjusted, and revised during reconstruction. The panorama is represented internally as a cube-map. User selections for insertion, cloning, or deletion are mapped to cubemap coordinates and extracted as image patches, with stitching applied when selections span multiple faces.

Within the panorama, users perform reconstruction through four edit operators that transform the scene state relative to this contextual representation. Insertion introduces new manipulable 3D cues when remembered elements are missing or underspecified, allowing users to externalize inferred details guided by the surrounding context. Movement adjusts object pose and depth to refine spatial relations when recalled cues are misaligned with the egocentric structure provided by the panorama. Cloning supports the externalization of contextual cues by extracting regions from the panorama background, removing surrounding context, and converting the result into reusable 3D assets that can be repositioned and reused. Deletion removes elements from the panorama background through localized editing and reintegration, treating suppression and correction of misleading contextual cues as first-class reconstruction actions rather than as visual cleanup. All generated objects are stored in an inventory for reuse across reconstructions.

Cloning supports two complementary workflows that enable efficient reuse of spatial cues. Users may duplicate any existing generated 3D object from the inventory for rapid reuse. Alternatively, users can clone an object that is part of the panorama background by selecting a cubemap region, extracting the corresponding patch, and applying image editing to remove background context and optionally normalize orientation before converting the result into a standalone 3D asset. Figure 3 illustrates the panorama cloning operation.

A combination of these operators supports layered edits within a single scene. For example, if a user wishes to place a new 3D cue in front of an object that is already present in the panorama background, they can first clone the obstructing background object into a 3D asset, delete it from the panorama, and then reposition the cloned object so that the newly inserted cue remains visually in front. This sequence illustrates how reconstruction proceeds through composed operations that incrementally resolve spatial uncertainty relative to the panoramic scaffold. Figure 4 illustrates the deletion operation in panorama mode, where a selected region of the 360 background is edited to remove an unwanted element and reintegrated into the panorama while preserving surrounding context.

3.2 Mode B: 3D grid for schematic, layout-first reconstruction

This mode provides a schematic reconstruction workflow as an alternative to the panorama-based approach. Users draw a layout on a grid, which is then instantiated as a 3D scene and populated with objects either selected from the inventory or created through

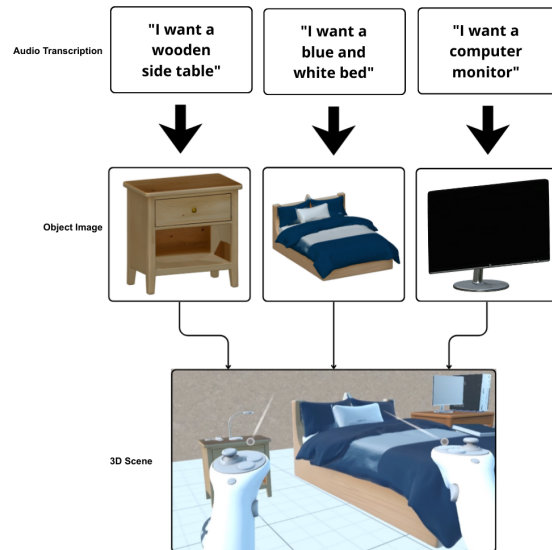


Figure 2: Example 3D reconstruction space populated through voice-driven object insertion. Generated object cues are instantiated as manipulable elements within a shared spatial scene, supporting placement, adjustment, and interaction during reconstruction.

3D generation. The resulting reconstruction emphasizes layout-first reasoning and supports a more allocentric style of spatial recall [4].

Because this mode does not rely on an underlying 360 background layer, panorama-specific operations such as background deletion and background cloning are not applicable. Instead, reconstruction focuses on explicit spatial structure, object placement, and relative layout, making this mode well suited for schematic or map-like representations of remembered environments. Figure 2 shows an example reconstruction space populated with multiple generated object cues, illustrating how voice-driven insertion integrates with spatial placement and interaction during scene reconstruction.

4 IMPLEMENTATION

The system maintains a scene state consisting of a background layer and a foreground set of object instances. In panorama mode, the background is represented as a cubemap that provides a stable contextual substrate. In grid mode, the background is an empty skybox and the user-defined grid is instantiated as a 3D layout.

Voice-driven operations use automatic speech recognition. The user records a short description, which is transcribed using Whisper and used as a text prompt for image generation or editing. For object insertion, the prompt is used to generate an object depiction suitable for 3D conversion. To reduce artifacts during 3D generation, we explicitly request uniform white backgrounds for generated object images.

An image-to-3D component then converts the generated depiction into a textured 3D asset, which is inserted into the scene as a manipulable object. Users can refine placement through direct manipulation, adjusting position, orientation, and depth relative to the panorama or grid. Text-to-3D generation is treated as enabling technology rather than the primary contribution [11]. Figure 5 illustrates the object generation pipeline, from spoken description to manipulable 3D asset.

Panorama deletion and background cloning are implemented as localized cubemap edits. For both operations, the user selects a region on the panorama, which is mapped to cubemap face coordinates and extracted as an image patch. When selections span multiple faces, extracted patches are stitched prior to editing. For

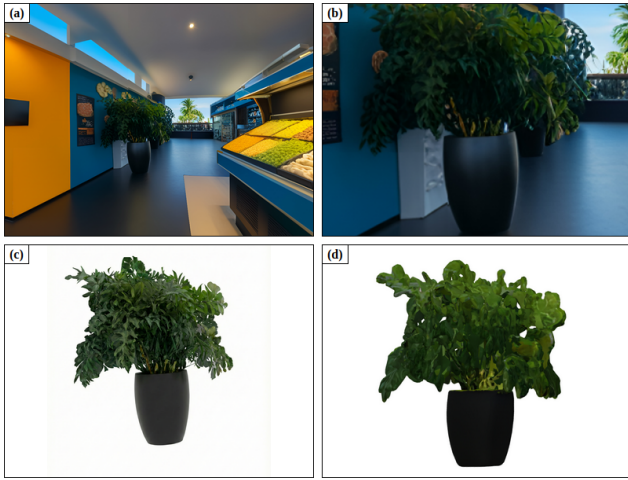


Figure 3: Panorama cloning. (a) Cubemap view containing the target object, (b) cropped image of the selected region, (c) background-removed object depiction, and (d) resulting 3D model used for reconstruction.



Figure 4: Panorama deletion through localized scene editing. A target element is selected from the panoramatic background, removed via instruction-guided editing, and the edited region is reintegrated into the scene.

deletion, instruction-guided editing or inpainting is applied to remove the target content, and the edited patch is reintegrated into the cubemap. For background cloning, the extracted patch containing the target object is processed to remove surrounding context and optionally normalize orientation before conversion into a 3D asset.

We initially explored segmentation-based removal for panorama edits, but localized crop-based editing proved more reliable for small corrections in the current pipeline. Treating deletion as a first-class operation reflects the view that reconstruction involves suppression and replacement of details, not only addition, and aligns with XR memory manipulation vocabularies emphasizing how XR systems can shape what is retained, forgotten, or distorted [2].

The current prototype uses generative models selected for their suitability to interactive reconstruction tasks. Skybox AI is used to generate initial 360 environments as contextual starting points. Image generation and editing, including object depiction, localized panorama edits, and cloning or removal operations, are handled by NanoBanana due to its instruction-following behavior and ability to make targeted adjustments while preserving surrounding context. For 3D object creation, Hunyuan3D 2.1 converts generated images into manipulable mesh assets that can be placed, moved, and reused during reconstruction. These components are modular, and the system design does not depend on model-specific features beyond localized editing and rapid iteration.

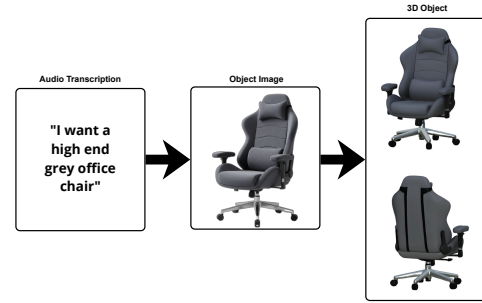


Figure 5: Voice-driven object generation as a reconstruction operator. A spoken description is transcribed and transformed into a manipulable 3D asset, enabling semantic-to-spatial insertion of new cues during scene reconstruction.

5 DISCUSSION

The system operationalizes constructive spatial memory reconstruction as iterative scene editing, enabling investigation of how people rebuild remembered environments through explicit manipulations of spatial cues. By separating representational scaffolds from reconstruction operators, the system provides a basis for future analyses that distinguish the effects of contextual structure from the effects of specific edit actions such as insertion, movement, cloning, or deletion.

The four reconstruction operators correspond closely to processes described in theories of constructive remembering. Insertion supports schema-guided completion when remembered scenes are underspecified [1]. Movement enables iterative refinement of spatial relations and transformations between egocentric and allocentric frames during imagery and recall [5, 4]. Cloning allows contextual elements that are implicitly present in memory to be externalized as explicit cues that can be repositioned and reused, supporting scene construction as described by Hassabis and Maguire [6]. Deletion reflects the suppression or correction of misleading details, consistent with accounts that view forgetting and distortion as adaptive components of memory rather than as failures [12]. Together, these operators operationalize reconstruction as a sequence of interpretable actions rather than as an opaque replay process.

The design emphasizes that reconstruction is not purely additive. Deletion and correction are treated as first-class operations, reflecting evidence that remembering involves suppressing irrelevant or misleading details as well as reinstating salient ones. This aligns with XRMemory's interest in memory manipulation as an intentional and interactive process rather than as a passive consequence of capture or replay. Making reconstruction actions explicit and traceable further enables examination of how particular edits may influence subsequent recall, confidence, or reinterpretation of remembered environments.

The reliance on generative components introduces known limitations. Image and panorama editing may introduce artifacts or plausible but incorrect content, and text-to-3D generation may produce stylized or inaccurate objects. These effects mirror well-documented characteristics of human memory, where reconstruction can yield coherent but imperfect representations [12]. Rather than assuming generative outputs are correct, the system is intentionally designed to support iterative adjustment, replacement, and removal of spatial cues. This design reflects a deliberate trade-off between flexibility and strict veridicality, prioritizing externalization of uncertainty and user-driven correction over automated fidelity.

6 CONCLUSION

We presented a VR system that frames spatial memory reconstruction as an iterative scene-editing process. A panorama-based scaffold reinstates egocentric context, while explicit operators for inserting, moving, cloning, and deleting objects support constructive rebuilding of remembered environments. A semantic-to-spatial pipeline enables rapid creation of manipulable 3D cues from voice descriptions, and localized panorama editing supports correction and suppression of contextual elements. The primary contribution is an interaction grammar that treats spatial memory as adaptive and editable rather than as passive replay.

Future work will consist of controlled user studies evaluating reconstruction quality and spatial accuracy under different representational scaffolds and operator sets. These studies will focus on behavioral and subjective measures appropriate for XRMemory, with the goal of understanding how explicit reconstruction actions shape remembered spatial scenes and how XR systems may influence the content and structure of memory through interaction.

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